

UDAY BHASKAR VALAPADASU

AI Engineer | MSCS – University of North Texas

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EXPERIENCE

Applied AI Engineer

Centific (Client: Citizens Bank)

February 2024 – Present

United States (Remote)

- Architected scalable data ingestion pipelines using Python, SQLAlchemy, and AWS Glue, processing millions of customer interaction logs and transaction histories in PostgreSQL with full data lineage for SOC 2 and PCI DSS audit readiness.
- Engineered behavioral features including intent classification, sentiment scoring, and transaction pattern analysis using pandas and spaCy, improving downstream model signal quality by **27%**.
- Fine-tuned transformer models (BERT, RoBERTa) on AWS SageMaker for intent detection and personalized financial product recommendations, increasing model accuracy from **0.78 to 0.90** through domain-specific tuning.
- Applied LoRA and PEFT via Hugging Face Accelerate on AWS EC2, enabling efficient fine-tuning on proprietary banking datasets under strict data privacy and governance standards.
- Ran 100+ hyperparameter optimization experiments using Optuna (Bayesian search, k-fold CV), tracking all results in MLflow with GitLab CI/CD integration for reproducible, automated model deployment.
- Deployed real-time conversational AI pipelines using LangChain, FAISS, and AWS Lambda to power intelligent banking chatbots, driving a **22% increase** in customer satisfaction scores.
- Built interactive analytics dashboards using Streamlit and Plotly on AWS Elastic Beanstalk, enabling real-time visualization of chatbot engagement metrics for product and customer success teams.

Graduate Teaching Assistant – Software Engineering

University of North Texas

August 2023 – December 2023

Denton, TX

- Supported a graduate-level Software Engineering course covering SDLC methodologies, system design, version control, testing strategies, and design patterns for 30+ students.
- Evaluated assignments, coding projects, and exams; delivered detailed feedback on code quality, documentation standards, and architecture decisions throughout the semester.

Full Stack Developer

EPAM Systems

February 2022 – June 2023

Remote

- Built high-performance frontend components in React, TypeScript, and Redux with real-time trade dashboards via WebSockets, improving trader productivity by **23%**.
- Designed Spring Boot microservices for trade validation, routing, and compliance checks; integrated Kafka event streams and FIX protocol adapters, reducing API latency by **27%**.
- Optimized PostgreSQL schemas, indexing strategies, and query plans for a Trade Order Management System processing millions of trades monthly, improving query performance by **31%**.
- Deployed containerized services on AWS ECS (Fargate) with autoscaling and AWS KMS key management, maintaining **99.99% uptime** while cutting infrastructure costs by **18%**.
- Built automated test pipelines using JUnit, Selenium, and GitHub Actions, raising test coverage to **92%** and reducing production defects across release cycles.

PROJECTS

B0Bot – Cybersecurity News Intelligence Platform | Open Source Contributor

Present

- Python, Flask, Apache Kafka, Debezium, Redis, Pinecone, LangChain, LangGraph, HuggingFace, SentenceTransformers, BeautifulSoup, httpx, PRAW*
- Engineered a production-grade cybersecurity intelligence platform aggregating threat data across 10 categories (malware, CVEs, data breaches, cyberattacks) from 15+ sources including The Hacker News, CISA, Palo Alto Unit 42, and GitHub Security Advisories.
- Designed a CDC pipeline using Apache Kafka with RSS readers and Debezium connectors – producer deduplicates articles by URL and publishes to **cybernews.raw**; consumer embeds via SentenceTransformer and upserts into Pinecone, eliminating costly full-index rebuilds.
- Built a multi-layer ingestion system combining concurrent web scraping (BeautifulSoup + httpx + ThreadPoolExecutor), 7 RSS feeds, YouTube Data API v3, NewsAPI.org, and social APIs (Reddit/PRAW, HackerNews Algolia, Mastodon, GitHub Advisories) with graceful credential-absent degradation.

- Implemented Redis caching for frequent news queries, reducing Pinecone round-trips and LLM inference overhead on repeated category lookups.
- Architected an agentic framework using LangChain/LangGraph with four specialized agents – Scraper, Analyzer, Responder, and Notification – enabling fully autonomous pipeline execution with planner-executor coordination.
- Extended HuggingFace InferenceClient to support multi-turn conversational dialogue, enabling follow-up queries and session-level context refinement beyond single-shot responses.
- Built a subscription system for daily/weekly email digests of curated threat summaries, delivered autonomously via the Notification Agent.
- Designed and implemented an improved real-time dashboard UI visualizing news streams, category breakdowns, and Kafka consumer lag metrics for operational observability.
- Wrote comprehensive API test suites across all connectors and pipeline stages with input validation, error handling, and edge case coverage.

DriveAssist AI | *Python, LangChain, FAISS, ChromaDB, BM25, AWS Lambda*
2026

- Built a RAG-based driving knowledge assistant using a hybrid retrieval pipeline combining ChromaDB vector store, BM25 sparse retrieval, and FAISS ANN search, improving answer relevance over single-method baselines.
- Orchestrated document chunking, embedding generation, and multi-step query chaining via LangChain for accurate, grounded responses from vehicle owner’s manual ingestion.
- Deployed as a serverless API on AWS Lambda, achieving scalable low-latency inference with zero idle infrastructure cost.

AssistMap - Accessible Restroom Locator | *Django, PostgreSQL, AWS EC2/RDS/SES, JS, Bootstrap*
2024

- Built a full-stack accessibility web app using Django and PostgreSQL enabling specially-challenged users to locate public restrooms via real-time geolocation search, distance calculation, and rating-based sorting.
- Implemented secure user management with custom profiles, email-verified registration, AWS SES password reset, and session-based favorites and search history.
- Developed an accessibility-focused blog module (“Area of Focus”) and responsive UI with interactive maps and photo galleries using Bootstrap, served across desktop and mobile.
- Deployed on AWS EC2 with RDS (PostgreSQL), Gunicorn, and WhiteNoise with environment-based config and production-grade static file handling.

TECHNICAL SKILLS

LLMs & NLP: Transformer Fine-Tuning (BERT, RoBERTa), LoRA, PEFT, LangChain, LangGraph, FAISS, ChromaDB, BM25, RAG, NER, Intent Classification, Sentiment Analysis, Hugging Face, SentenceTransformers, spaCy, Tokenization

ML & Deep Learning: PyTorch, TensorFlow, Scikit-learn, XGBoost, LightGBM, CNNs, LSTMs, GANs, Optuna, Bayesian Optimization, MLflow

Cloud & MLOps: AWS SageMaker, Lambda, EC2, ECS (Fargate), Glue, RDS, SES, KMS, Elastic Beanstalk, Azure ML, GCP AI Platform, Docker, Kubernetes, GitLab CI/CD, GitHub Actions

Data Engineering: Apache Kafka, Debezium, PySpark, Apache Airflow, Azure Data Factory, PostgreSQL, SQLAlchemy, pandas, NumPy, BeautifulSoup, httpx, PRAW, Pydantic, REST APIs, Web Scraping

Backend & Frameworks: Python, Flask, Django, FastAPI, Spring Boot, Node.js, Supabase, Gunicorn, Microservices, FIX Protocol

Frontend & Visualization: React, TypeScript, Redux, WebSockets, JavaScript, HTML/CSS, Bootstrap, Vite, Streamlit, Plotly, Matplotlib

Databases & Caching: PostgreSQL, Redis, Pinecone (Vector DB), ChromaDB

Monitoring & Explainability: SHAP, LIME, Evidently, Prometheus, Grafana, CloudWatch

Compliance & Security: SOC 2, PCI DSS, Data Lineage, AWS KMS, Data Privacy Governance

EDUCATION

University of North Texas	Denton, TX
<i>Master of Science in Computer Science</i>	<i>August 2023 – December 2024</i>
Sphoorthy Engineering College	India
<i>Bachelor of Technology in Computer Science and Engineering</i>	<i>2018 – 2022</i>

CERTIFICATIONS

AWS Certified AI Cloud Practitioner | Docker Professional Certificate